

Poisson pr.

$$\{N(t) : t \geq 0\}.$$

$$- N(0) = 0$$

- Has st. incr.

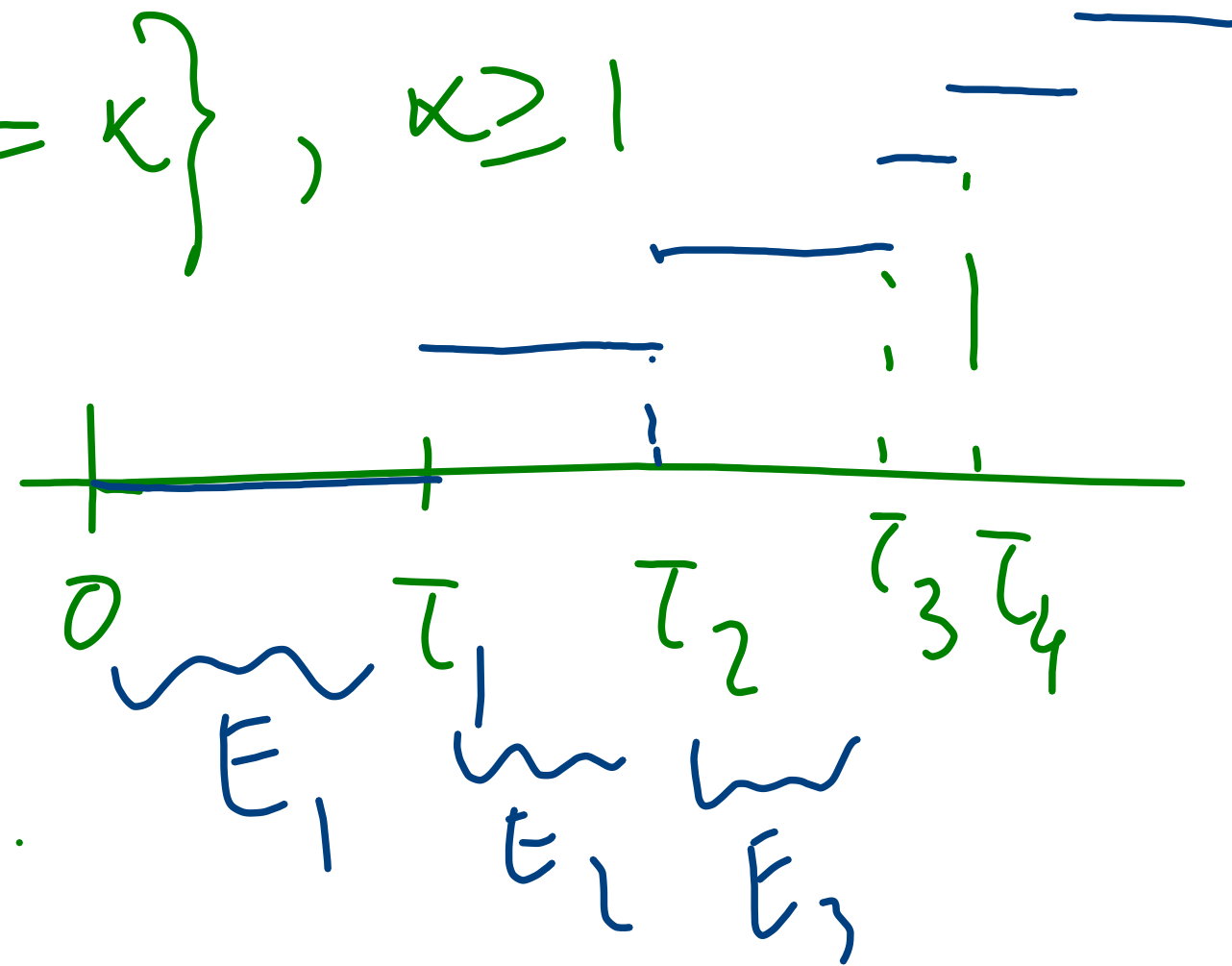
$$- P(N(t) = k) = \frac{e^{-\lambda t} (\lambda t)^k}{k!}, k \geq 0.$$

$$\begin{array}{l} P(N(t+h) - N(t) = i) \\ \quad = \lambda h + o(h) \\ P(N(t+h) - N(t) \geq 2) \\ \quad = o(h). \end{array}$$

Inter arrival times

$$T_0 = 0, \quad T_k = \inf\{N(t) = k\}, \quad k \geq 1$$

Inter arrival times

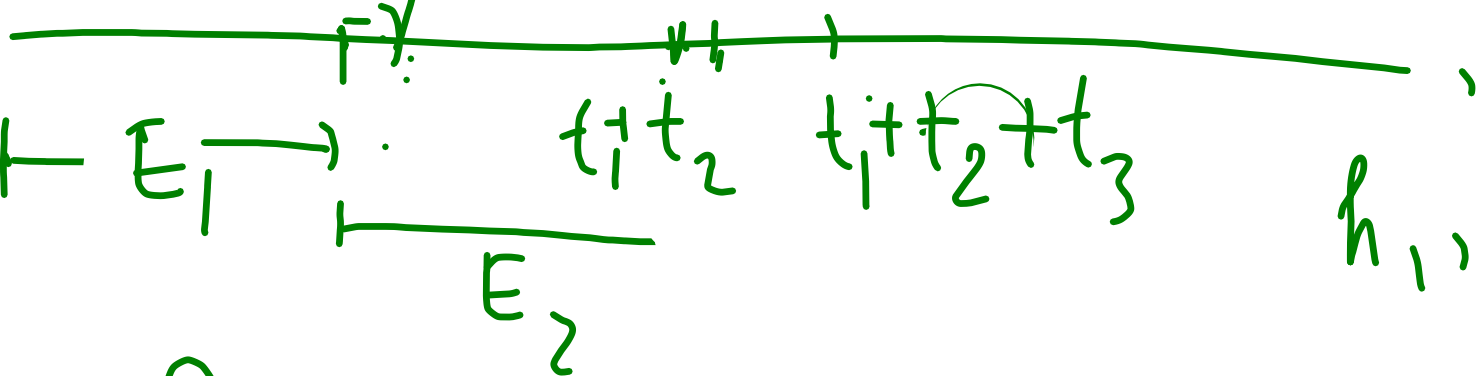


$$E_k = T_k - T_{k-1}, \quad k \geq 1.$$

Propⁿ:- $\{E_i : i \geq 1\}$ iid $\text{Exp}(\lambda)$.

$$f_{x_1, x_2}(x_1, x_2) = \lim_{h_1, h_2 \rightarrow 0} \frac{P\left(\begin{array}{l} x_1 \in (x_1, x_1+h_1] \\ x_2 \in (x_2, x_2+h_2] \end{array}\right)}{h_1 h_2}$$

Fix $t_1, t_2, \dots, t_k > 0$



$$f_{E_1, E_2, \dots, E_k}(t_1, t_2, \dots, t_k) = \lim_{h_1, h_2, \dots, h_k > 0} \frac{P\left(\begin{array}{l} t_i \in (t_i, t_i+h_i] \\ t_i+h_i < t_{i+1} \end{array}\right)}{h_1 h_2 \dots h_k}$$

(choose h_i s.t. $t_i+h_i < t_{i+1}$)

Fix k , and $t_1, \dots, t_k$, and $h_1, \dots, h_k$
 such that $t_i + h_i < t_{i+1}$, for $i=1, 2, \dots, k-1$.

$$P(E_1 \in (t_1, t_1 + h_1), E_2 \in (t_2, t_2 + h_2), \dots, \\ E_k \in (t_k, t_k + h_k))$$

$$= P(N(t_1) = 0, N(t_1 + h_1) - N(t_1) = 1, N(t_1 + t_2) - N(t_1 + h_1) \\ N(t_1 + t_2 + h_1) - N(t_1 + t_2) = 1, \dots, N(t_1 + t_2 + \dots + t_{k-1} + t_k) \\ N(t_1 + \dots + t_k + h_k) - N(t_1 + \dots + t_k) = 1, N(t_1 + t_2 + \dots + t_{k-1} + t_k) = 0,$$

$$= P(N(t_1) = 0) P(N(t_1 + h_1) - N(t_1) = 1) \checkmark$$

$$P(N(t_1 + t_2) - N(t_1 + h_1) = 0) P(N(t_1 + t_2 + h_2) - N(t_1 + t_2) = 1) \checkmark$$

$$\vdots$$

$$P(N(t_1 + t_2 + \dots + t_k) - N(t_1 + \dots + t_{k-1} + h_{k-1}) = 0) \checkmark$$

$$P(N(t_1 + \dots + t_k + h_k) - N(t_1 + \dots + t_k) = 1)$$

$$= \prod_{i=1}^k P(N(h_i) = 1) e^{-\lambda t_1} \prod_{i=2}^k e^{-\lambda(t_i - h_{i-1})}$$

$$= \prod_{i=1}^k e^{-\lambda h_i} (\lambda h_i) \times \swarrow$$

$$= \lambda^k \prod_{i=1}^k h_i e^{-\lambda \sum_{i=1}^k h_i} e^{-\lambda \sum_{i=1}^k t_i} e^{\lambda \sum_{i=1}^{k-1} h_i}$$

$$= \lambda^k \prod_{i=1}^k h_i e^{-\lambda \sum_{i=1}^k t_i} e^{-\lambda h_k}$$

The density of $E_1, \dots, E_k$ is given by

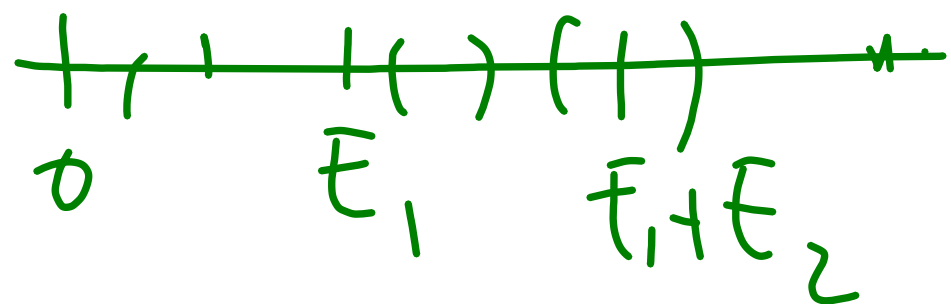
$$\lim_{h_1, \dots, h_k \rightarrow 0} \frac{1}{\prod_{i=1}^k h_i} \times \lambda^k \prod_{i=1}^k h_i e^{-\lambda \sum_{i=1}^k t_i} e^{-\lambda h_k} = \prod_{i=1}^k [\lambda e^{-\lambda t_i}]$$

E_i iid, $S_0=0$, $S_n = \sum_{i=1}^n E_i$, $E_i \sim \exp(\lambda)$.

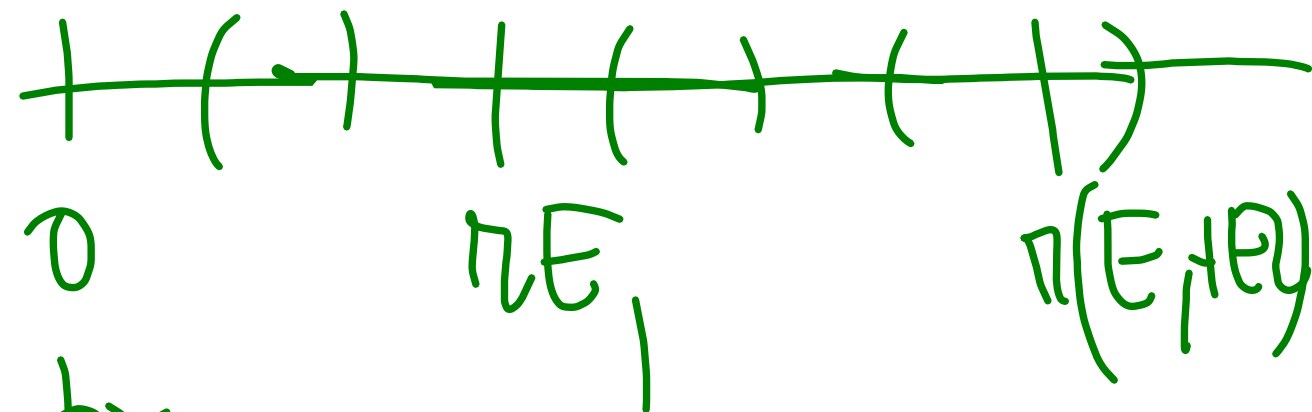
Define $N(t) = \max\{n : S_n \leq t\}$.

— $N(t)$ is a Poisson pr.

Scaling $\therefore \lambda > 0$.



$\Rightarrow$



$N_\lambda(t)$ is also a Poisson pr.

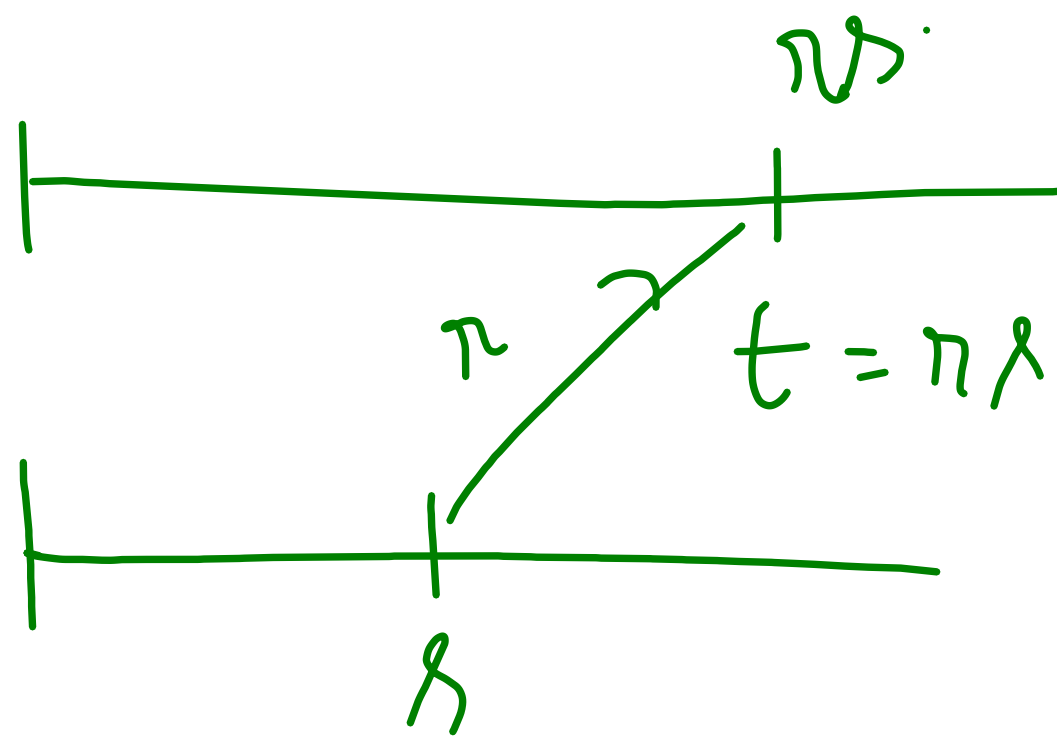
$$P(N_{\eta}(t) = k).$$

$$= P(N_{\eta}(\eta s) = k).$$

$$= P(N(s) = k).$$

$$= e^{-\lambda s} \frac{(\lambda s)^k}{k!} = e^{-\lambda t/\eta} \frac{(\lambda t/\eta)^k}{k!} = e^{-\lambda_{\eta} t} \frac{(\lambda_{\eta} t)^k}{k!}$$

$$\lambda_{\eta} = \frac{\lambda}{\eta}.$$



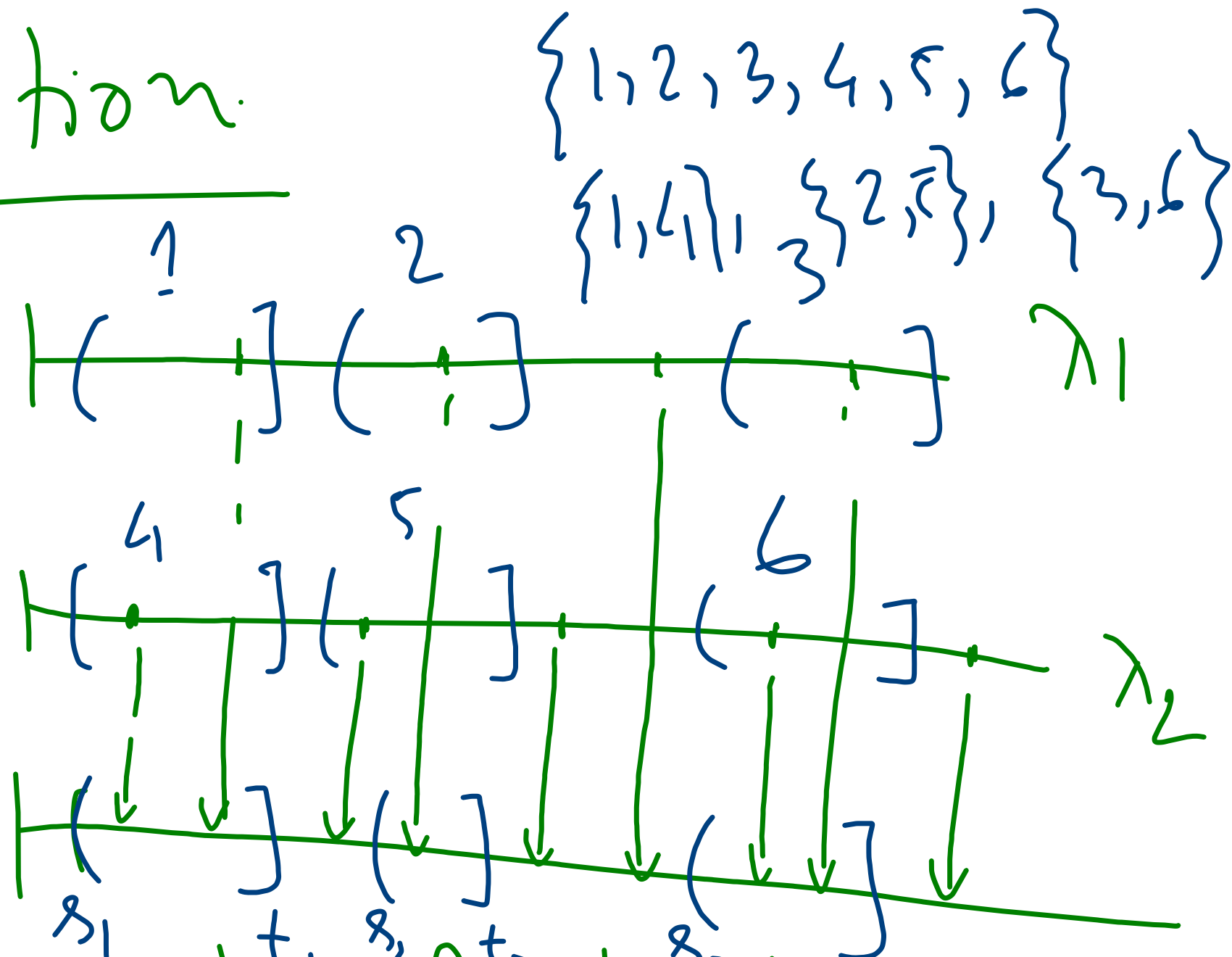
Superposition.

indep $\begin{cases} N_1(t) \\ N_2(t) \end{cases}$

$$N^+(t) = N_1(t) + N_2(t)$$

N^+ is a Poisson pr. of intensity $\lambda_1 + \lambda_2$.

$\{N_1(t_i) - N_1(s_i) : i=1, \dots, k, N_2(t_i) - N_2(s_i) : i=1, \dots, k\}$ are indep



$$N^+(t) = N_1(t) + N_2(t) \sim \text{Poi}((\lambda_1 + \lambda_2)t)$$

$\downarrow$
 $\text{Poi}(\lambda_1 t)$

$\downarrow$
 $\text{Poi}(\lambda_2 t)$

Conditional distoⁿ.

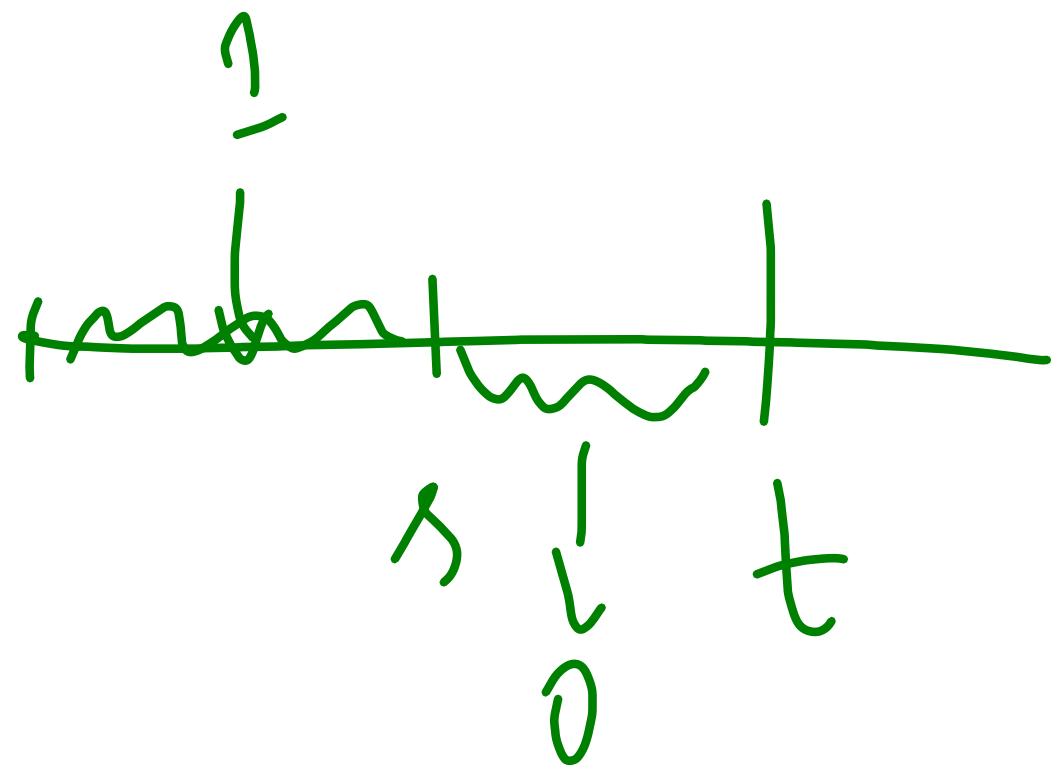
$$N(t) = 1.$$

$$P(\tau_1 \leq s \mid N(t) = 1).$$

$$= P(\tau_1 \leq s, N(t) = 1) / P(N(t) = 1)$$

$$= P(N(s) = 1, N(t) = 1) / \checkmark.$$

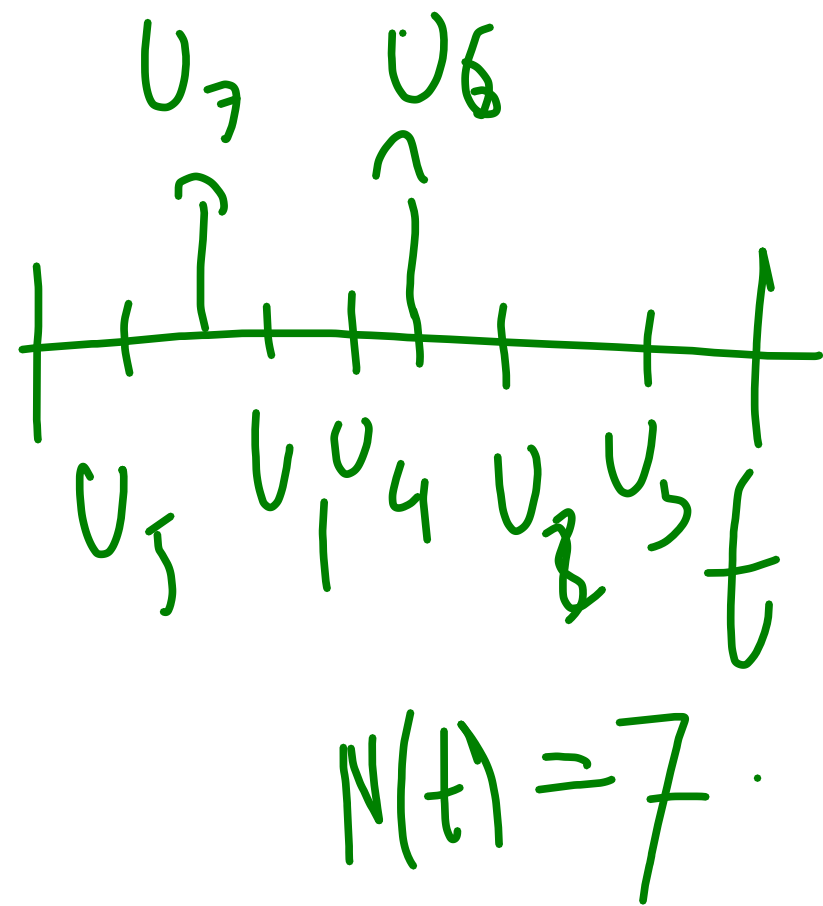
$$= P(N(s) = 1, N(t) - N(s) = 0) / \checkmark$$



$$= \frac{e^{-\lambda s} \cdot \lambda s \cdot e^{-\lambda(t-s)}}{e^{-\lambda t} \cdot \lambda t}$$

$$= \frac{s}{t} = P(U(0,t) \leq s)$$

$$\tau_1 \mid N(t) = 1 \sim U(0,t)$$



$$f_{T_1, \dots, T_k | N(t)=k} (s_1, s_2, \dots, s_k)$$

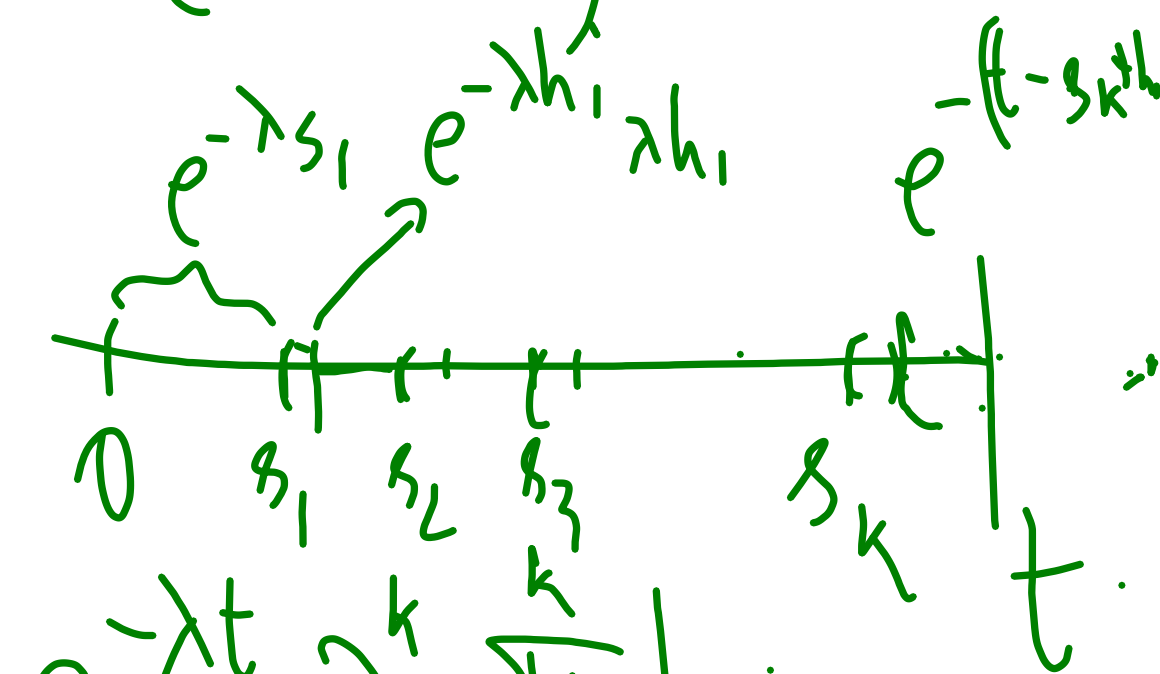
$$\underline{s_1 < s_2 < \dots < s_k \leq t.}$$

$$P(T_i \in (s_i, s_i + h_i), i=1, \dots, k, N(t)=k)$$

$$= e^{-\lambda t} \cdot \lambda^k \prod_{i=1}^k h_i$$

The conditional density

$$f_{T_1, \dots, T_k | N(t)=k} (s_1, \dots, s_k) = \lim_{h_1, \dots, h_k \rightarrow 0} \frac{k!}{t^k}$$



$$e^{-\lambda t} \lambda^k \prod_{i=1}^k h_i$$

$$\frac{\prod_{i=1}^k h_i e^{-\lambda t} (\lambda t)^k / k!}{\prod_{i=1}^k h_i e^{-\lambda t} (\lambda t)^k / k!}$$

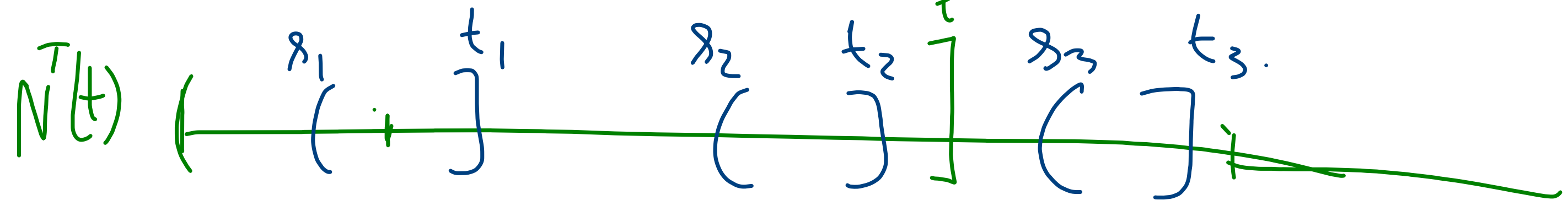
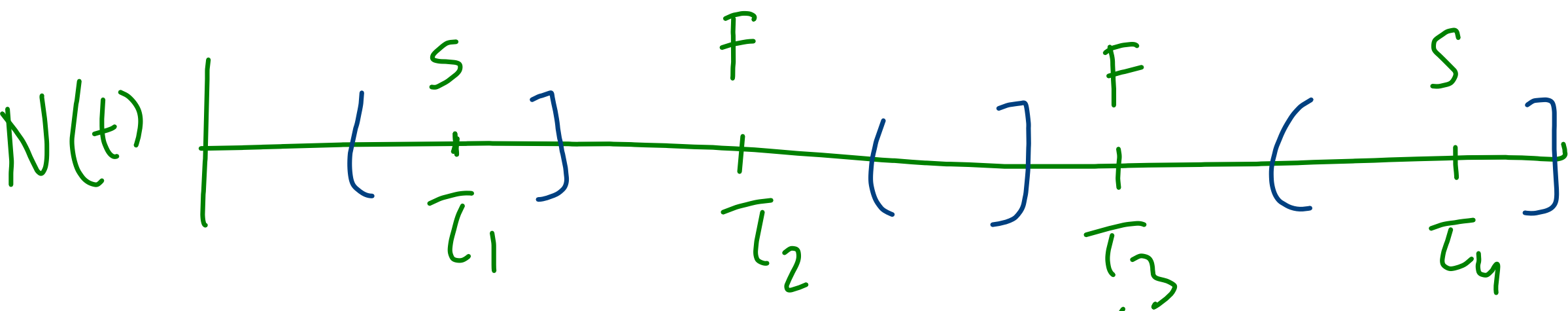
Let $U_1, U_2, \dots, U_k$ iid $U(0, t)$.

Consider the order statistic

$$U_{(1)}, U_{(2)}, U_{(3)} \dots, U_{(k)}$$

$$f_{U_{(1)}, U_{(2)}, \dots, U_{(k)}}(s_1, s_2, \dots, s_k) = \begin{cases} \frac{k!}{t^k} & \text{if } 0 < s_1 < s_2 < \dots < s_k < t \\ 0 & \text{o.w.} \end{cases}$$

Therefore the conditional distⁿ of $T_1, \dots, T_k$ is the same as the order statistic from uniform $U(0, t)$.



$$\{N^T(t_1) - N^T(s_1), N^T(t_2) - N^T(s_2), N^T(t_3) - N^T(s_3)\}$$

$$P(N^T(t) = k) = \sum_{n=k}^{\infty} P(\cancel{N^T(t) = k}, N(t) = n, \text{out of } n \text{ arrivals, } k \text{ many suc.})$$

$$= \sum_{n=k}^{\infty} \frac{e^{-\lambda t} (\lambda t)^n}{n!} \cdot \binom{n}{k} p^k (1-p)^{n-k}.$$

$$= \frac{e^{-\lambda t} p^k}{k!} \sum_{n=k}^{\infty} \frac{(\lambda t)^k}{(\lambda t)^k} \cdot \frac{n!}{(n-k)! k!} (1-p)^{n-k}.$$

$$= \frac{e^{-\lambda t} (\lambda p t)^k}{k!} \sum_{n=k}^{\infty} \frac{(\lambda t (1-p))^{n-k}}{(n-k)!} \cdot \lambda t (1-p)$$

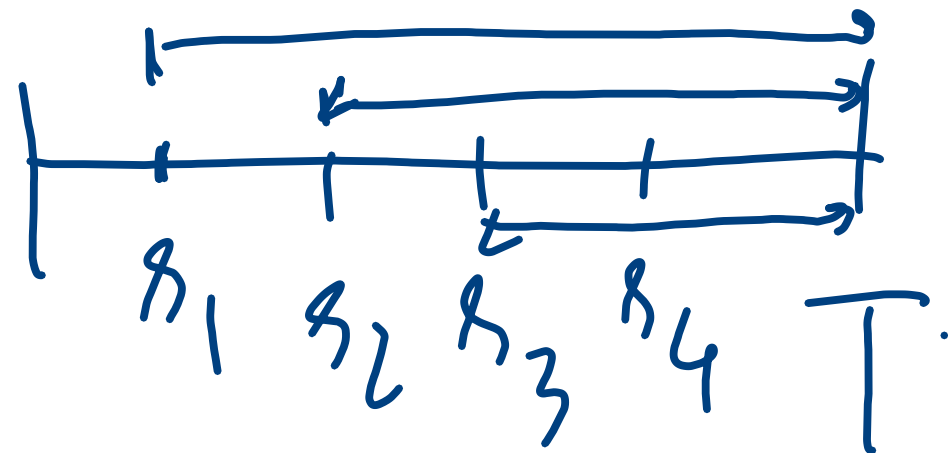
$$= \frac{e^{-\lambda t} (\lambda p t)^k}{k!} e^{\lambda t (1-p)} = e^{-\lambda p t} \cdot (\lambda p t)^k / k!$$

For fixed T ,
 The total accumulated

waiting time

$$W = \sum_{k=1}^{N(T)} (T - \tau_k)$$

$$E\left(\sum_{k=1}^{N(T)} (T - \tau_k)\right) = E\left(E(W | N(T))\right)$$



Given
 $N(T) = k$.

$$= \sum_{i=1}^k (T - \tau_i)$$

$$= kT - \sum_{i=1}^k \tau_i$$

$$\stackrel{d}{=} kT - \sum_{i=1}^k U_{(i)}$$

$$= kT - \sum_{i=1}^k U_i$$

$$U_{(1)} + U_{(2)} + \dots + U_{(k)} \\ = \sum_{i=1}^k U_i$$

$U_1, \dots, U_k \sim \text{uniform}$
 $U(0, T)$
 $U_{(1)}, \dots, U_{(k)}$

$$E(W | N(T) = k)$$

$$= kT - E\left(\sum_{i=1}^k U_i\right)$$

$$= kT - \sum_{i=1}^k E(U_i)$$

$$= kT - kT/2 = kT/2$$

So, $E(W | N(T)) = N(T) T / 2$

$$E(W)$$

$$= E(E(W | N(T)))$$

$$= E\left(\frac{T N(T)}{2}\right)$$

$$= \frac{1}{2} \lambda T$$

$$= \frac{\lambda T^2}{2}$$